Lab 3: Operational Amplifiers

- Review
  - Operational amplifiers (Op Amps)
    - Assumptions
      - $v^- = v^+$
      - $i_{in} = 0$
    - Design recommendations
      - Use resistors in the range of 1kΩ to 700 kΩ
      - (See also Rizzoni, page 550)
  - Model circuits with Op Amps
    - Null offset
    - Comparator
    - Follower
    - Inverting amplifier
    - Non inverting amplifier
    - Differential amplifier
    - Summing amplifier
    - Integrator
    - Differentiator
  - Resistor color band
    - (link)
  - Voltage divider
- Equipment
  - Breadboard with triple power supply
  - Sweep/function generator (link)
  - Oscilloscope (link)
  - 741 op amp (link)
    - Pin 1: Offset N1
    - Pin 2: v^-
    - Pin 3: v^+
    - Pin 4: V_S^- (can also connect a .1μF capacitor to ground)
    - Pin 5: Offset N2
    - Pin 6: v_{out}

    - Pin 7: V_S^+ (can also connect a .1μF capacitor to ground)
    - Pin 8: No internal connection
  - resistors: 1kΩ, 10kΩ
  - Capacitors: 10 nF

- **Overall, your objectives are to (i) identify and quantify major differences between the behavior of a 741 op amp and an ideal operational amplifier and (ii) develop and implement strategies to avoid/mitigate these differences in several common circuits.**
  - Consider addressing the following questions for each circuit you build and measure.
    - How do your measurements differ from the expected output for an ideal operational amplifier? Quantify the differences?
    - How do you anticipate using the circuit throughout the remainder of the semester? Considering these applications, would the differences you identified be significant?
    - What conditions or modifications could be put in place to avoid/mitigate these differences? Be specific.
  - Consider the following common circuits involving operational amplifiers
    1. Output with grounded inputs
       - Setup
         - $V_S^+ = 6$ V
         - $V_S^- = -6$ V
         - v^- : ground
         - v^+ : ground
         - v_{out} : measure
    2. Comparator
       - Setup
         - $V_S^+ = 6$ V
         - $V_S^- = -6$ V
         - v^- : ground (or a second input)
         - v^+ : input signal (try various functions and frequencies; e.g., sine, 500 $\mathrm{mV_{pp}}$, 1kHz)
         - v_{out} : measure
       - Notes
         - Record both the input and output of the circuit
    3. Follower

- ∗ Setup
  - · $V_S^+ = 6$ V
  - · $V_S^- = -6$ V
  - · v^- : connected to v_{out}
  - · v^+ : input signal (try various functions and frequencies; e.g., sine, 500 $\mathrm{mV_{pp}}$, 1kHz)
  - · v_{out} : measure (and connected to v^-)
- ∗ Notes
  - · Record both the input and output of the circuit

4. Inverting amplifier
   - ∗ Setup
     - · $V_S^+ = 6$ V
     - · $V_S^- = -6$ V
     - · v^- : connected to (i) input (try various functions and frequencies; e.g., sine, 500 $\mathrm{mV_{pp}}$, 1kHz) through a 1kΩ resistor and (ii) output through a 10 kΩ resistor
     - · v^+ : ground
     - · v_{out} : measure (and connected to v^- through a 10 kΩ resistor)
   - ∗ Notes
     - · You might want to measure the resistance of the resistors to assist in your analysis.
     - · Record both the input and output of the circuit
5. Noninverting amplifier
   - ∗ Setup
     - · $V_S^+ = 6$ V
     - · $V_S^- = -6$ V
     - · v^- connected to: (i) ground through a 1 kΩ resistor and (ii) output through a 10 kΩ resistor
     - · v^+ connected to input (sine, 500 mV_{pp}, 1 kHz) through 10kΩ resistor
     - · v_{out} measure (and connected to v^- through a 10 kΩ resistor)
     - · Is the measured V_{pp} what you expect considering the input? Attempt to explain any differences.
   - ∗ Notes
     - · You might want to measure the resistance of the resistors to assist in your analysis.
     - · Record both the input and output of the circuit
6. Differential amplifier

- Setup
  - $V_S^+ = 6$ V
  - $V_S^- = -6$ V
  - v^- : connected to (i) input (sine, 500 mV$_{pp}$, 1 kHz, **with 455mV DC offset**) through a 1kΩ resistor and (ii) output through a 10 kΩ resistor
  - v^+ : connected to (i) ground through 10 kΩ resistor and (ii) a 455 mV source through a 1kΩ resistor
  - v_{out} : measure (and connected to v^- through a 10 kΩ resistor)
- Notes
  - You might want to measure the resistance of the resistors to assist in your analysis.
  - Record the **two** inputs and the output of the circuit
  - The second input could be obtained from a voltage divider paired with a follower

7. Summing amplifier
   - Setup
     - $V_S^+ = 6$ V
     - $V_S^- = -6$ V
     - v^- : connected to (i) input (sine, 500 mV$_{pp}$, 1 kHz) through a 1kΩ resistor, (ii) input (DC, 455 mV) through a 1kΩ resistor, and (iii) output through a 1 kΩ resistor (Hint, use a voltage divider together with a follower circuits to obtain the 455mV input.)
     - v^+ : ground
     - v_{out} : measure (and connected to v^- through a 1 kΩ resistor)
   - Notes
     - You might want to measure the resistance of the resistors to assist in your analysis.
     - Record the **two** inputs and the output of the circuit
     - The second input could be obtained from a voltage divider paired with a follower
8. Integrator
   - Setup
     - $V_S^+ = 6$ V
     - $V_S^- = -6$ V
     - v^- : connected to (i) input (square, 500 mV$_{pp}$, 1 kHz) through a 10 kΩ resistor and (ii) output through a 10 nF capacitor

  - v^+ : ground
  - v_{out} :measure (and connected to v^- through a 10 nF capacitor)
- ∗ Notes
  - You might want to measure the resistance/capacitance of the resistors/capacitors to assist in your analysis.
  - Record both the input and the output of the circuit
  - Design/implement/test a method to improve performance of your circuit.

9. Differentiator
   - ∗ Setup
     - $V_S^+ = 6$ V
     - $V_S^- = -6$ V
     - v^- : connected to (i) input (sine, 500 mV$_{pp}$, 1 kHz) through a 10 nF capacitor and (ii) output through a 10 kΩ resistor
     - v^+ : ground
     - v_{out} : measure (and connected to v^- through a 10 kΩ resistor)
   - ∗ Notes
     - You might want to measure the resistance/capacitance of the resistors/capacitors to assist in your analysis.
     - Record both the input and the output of the circuit
     - Design/implement/test a method to improve performance of your circuit.

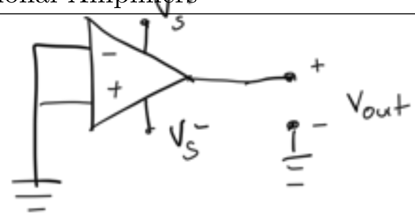
- Lab report:
  - Your lab report should follow the required lab report structure for this class
  - In the body of your report, you should build a strong case for your objectives: to (i) identify and quantify major differences between the behavior of a 741 op amp and an ideal operational amplifier and (ii) propose strategies to avoid/mitigate these differences in several common circuits.
  - Include the following in the Appendix:
    - ∗ This lab outline (required)
    - ∗ Data you collected during lab (required)
    - ∗ Responses to the following questions for each circuit you built in the lab
      - How do your measurements differ from the expected output for an ideal operational amplifier? Quantify the differences?
      - How do you anticipate using the circuit throughout the remainder of the semester? Considering these applications, would the differences you identified be significant?

    - What conditions or modifications could be put in place to avoid/mitigate these differences? Be specific.
  - Any other details (data, derivations, calculations, experimental procedure, screen captures etc.)

(The purpose of the Appendix in these reports is to: (i) serve as somewhat of a lab notebook, (ii) demonstrate to the grader what you did during lab, and (iii) hold material that didn't fit into the body of the report but would be necessary to repeat the experiments or justify the claims in the report.)

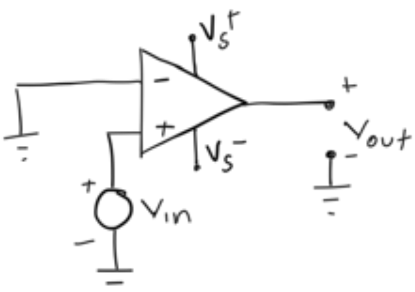
Null offset

See "Principles and Applications of Electrical Engineering," Rizzoni McGraw Hill 2000

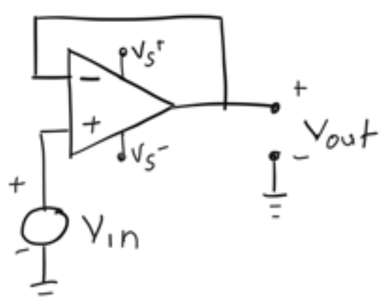


Comparator

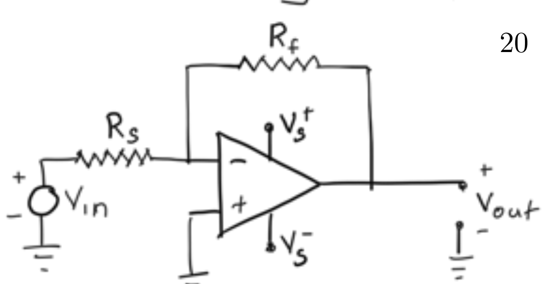
this comparator compares to ground.



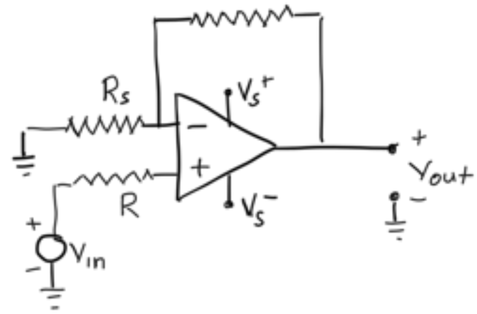
Follower



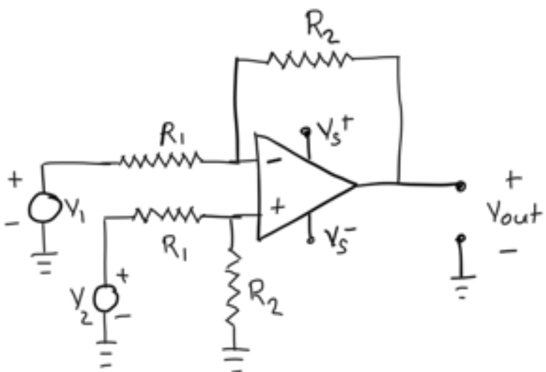
Inverting Amplifier



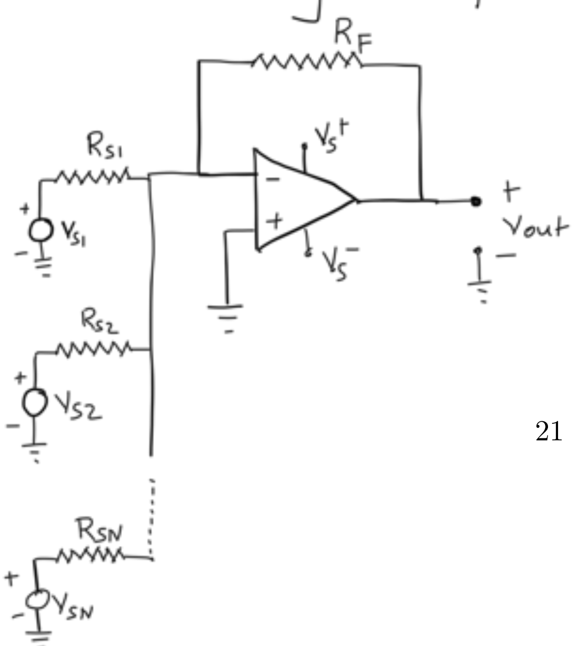
Noninverting Amplifier



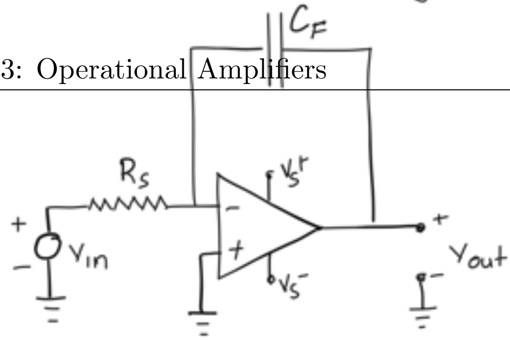
Differential Amplifier



Summing Amplifier



Ideal Integrator



Ideal Differentiator